

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination – 2023

Course: B. Tech. Branch: Civil & Allied Engineering Semester: III

Subject Code & Name: BTCIC305/BTCEC305/BTCVC305 Surveying

Max Marks: 60

Date: 11/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define surveying. Write down applications of surveying. State Principle of Surveying.	CO1	6
B) Explain different methods of linear measurement.	CO1	6
C) Differentiate between plane surveying and geodetic surveying.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain the various accessories used in plane table surveying with neat labelled diagram.	CO2	6
B) Define the following: (any three) a. Open traverse and closed traverse b. Fore bearing and Back bearing c. True meridian and Magnetic Meridian d. True bearing and Magnetic meridian	CO2	6
C) Convert the following WCB's to QB's a. WCB of AB = $45^{\circ}30'$ b. WCB of CD = $108^{\circ}45'$ c. WCB of GH = $75^{\circ}50'$ d. WCB of HK = $145^{\circ}20'$ e. WCB of ML = $340^{\circ}10'$ f. WCB of DE = $60^{\circ}10'$	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Write down the characteristics of contour lines.	CO3	6
B) The following staff readings were taken successfully with level, the instrument having been move forward after the second, fourth and eighth reading:	CO3	6

0.0875, 1.235, 2.310, 1.385, 2.930, 3.125, 4.125, 0.120, 1.875, 2.030, 3.765

The first reading was taken on a benchmark of elevation 132.135 Enter the reading in level book form and reduced level. Calculate the reduced level by rise and fall method.

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| C) Enlist the various types of leveling. Explain any one with neat labelled diagram. | CO3 | 6 |
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Q.4 Solve Any Two of the following. **12**

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| A) State the functions of following parts of Theodolite: 1. Eyepiece 2. Bubble tube 3. Lower Tangent screw 4. Index Frame 5. Telescope | CO4 | 6 |
| B) What are the different methods of traversing? Explain any one in detail. | CO4 | 6 |
| C) Define (any three): | CO4 | 6 |
| 1. Transiting of Theodolite | 4. Telescope Normal | |
| 2. Swinging | 5. Face left | |
| 3. Line of Collimation | | |

Q. 5 Solve Any Two of the following. **12**

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|---|-----|---|
| A) Explain project survey on Railway Engineering. | CO4 | 6 |
| B) Write a Short note on Mine Surveying. | CO4 | 6 |
| C) Explain the steps involved while carrying out Engineering Survey | CO1 | 6 |

*** End ***